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(Affiliated to VTU, Belagavi)

JP Nagar 8th Phase, Bengaluru - 560076

Department of Mechanical Engineering

RENEWABLE ENERGY POWER PLANTS

(BME654B)



Course Name: RENEWABLE ENERGY POWER PLANTS

Course Code: BME654B

VI Semester

2025 Scheme

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- **PSO2:** Proficient to use all appropriate software to solve complex engineering problems.
- **PSO3:** Impart quality knowledge in the field of management and interdisciplinary technologies to suit industry and society.

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Chapter 1

Wind and Geothermal Energy Conversion

Wind Power Plants: Basics of wind energy and wind turbine technology, Types of wind turbines: horizontal axis and vertical axis; Wind resource assessment and site selection for wind power plants, Wind farm layout optimization and wake effects, Grid integration and power system considerations for wind power plants

Geothermal Energy Conversion: Principle of working, types of geothermal station with schematic diagram, geothermal plants in the world, problems associated with geothermal conversion, scope of geothermal energy.

1.1 Introduction

Wind is caused by the uneven heating of the atmosphere by the sun, variations in the earth's surface, and rotation of the earth. Thus the wind energy is a form of solar energy

Wind is the moving air and is caused by the differences in air pressure in our atmosphere. Mountains, bodies of water, and vegetation all influence wind flow patterns

Wind energy (or wind power) describes the process by which wind is used to generate electricity

Wind turbines convert the kinetic energy in the wind into mechanical power by rotating propeller like blades. A generator can convert mechanical power into electricity i.e. by rotating the propeller blades around a rotor. Mechanical power can also be utilized directly for specific tasks such as pumping water.



1.2 Properties Of Wind

- Wind is non-conventional energy source
- Wind is due to differences in air pressure in the atmosphere
- Wind at high pressure tends to move to areas at low air pressure, greater the pressure difference faster will be the flow of air
- In meteorology, winds are often referred to according to their strength, and the direction from which the wind is blowing.
- Wind strength can vary from light breeze to hurricane force
- The wind is also a critical means of transportation for seeds, insects, and birds, which can travel on wind currents for thousands of miles.
- Wind is characterized by two parameters. They are wind speed and wind direction. Wind speed indicates the speed of air movement from one point to another measured through a device called “anemometer” and the wind direction indicated the direction at which air is moving measured through a device “wind vane” attached to a direction indicator.

1.3 Availability Of Wind Energy In India

The total installed capacity of wind power in India as on March 2017 is around 32 GW. Wind power generation capacity in India has significantly increased in recent years. As of 28 February 2021, the total installed wind power capacity is 38.789 GW, the fourth largest installed wind power capacity in the world.

Wind power capacity is mainly spread across the Southern, Western and Northern regions (Note: GW = Gigawatt, 1 GW = 1000 megawatt = 10^9 watts)

Wind power costs in India are decreasing rapidly. The levelised tariff of wind power reached a record low of Rs. 2.43 per kWh (without any direct or indirect subsidies) during auctions for wind projects in December 2017. However, the levelised tariff is increased to Rs. 2.77 per kWh in March 2021.

The potential is far from exhausted. Indian Wind Energy Association has estimated that with the current level of technology, the 'on-shore' potential for utilization of wind energy for electricity generation is of the order of 65,000 MW. The unexploited resource availability has the potential to sustain the growth of wind energy sector in India in the years to come.

Wind in India are influenced by the strong south-west summer monsoon, which starts in May-June, when cool, humid air moves towards the land; further, the weak north-east winter monsoon, which starts in October, when cool, dry air moves towards the ocean. During March- August, the winds are uniformly strong over the whole Indian Peninsula, except the eastern peninsular coast. Wind speeds during November-March are relatively weak, although high winds are available during a part of the period on the Tamil Nadu coastline.

1.4 Wind Velocity

Wind velocity is the measure of speed of wind in horizontal direction. Wind velocity means pedestrian level wind speed measured at 2 m above ground. It is a measure of air ventilation which has a direct effect on outdoor thermal comfort.

1.5 Wind Power

Kinetic energy exists whenever an object of a given mass is in motion with a translational or rotational speed. When air is in motion, the kinetic energy in moving air can be determined as

$$E_k = \frac{1}{2} \times m \times u^2 \quad (1.1)$$

Where m is the air mass and u is the mean wind speed over a suitable time period.

The wind power can be obtained by differentiating the kinetic energy in wind with respect to time, i.e.

$$P = \frac{dE_k}{dt} = \frac{1}{2} \times m \times u^2 \quad (1.2)$$

However, only a small portion of wind power can be converted into electrical power.

When wind passes through a wind turbine and drives blades to rotate, the corresponding wind mass flow rate is

$$m = \rho \times A \times u^2 \quad (1.3)$$

Where ρ is the air density and A is the swept area of blades

Substituting (3) into (2), the available power in wind P_w can be expressed as

$$P_w = \rho \times A \times u^3 \quad (1.4)$$

1.6 Major Problems Associated With Wind Power

Wind energy can have adverse environmental impacts, including the potential to reduce, fragment, or degrade habitat for wildlife, fish, and plants.

- Wind power must still compete with conventional generation sources on a cost basis i.e. wind projects must be able to compete economically with the lowest-cost source of electricity, and some locations may not be windy enough to be cost competitive.
- Good land-based wind sites are often located in remote locations, far from cities where the electricity is needed. Transmission lines must be built to bring the electricity from the wind farm to the city.
- Wind resource development might not be the most profitable use of the land. Land suitable for wind- turbine installation must compete with alternative uses for the land, which might be more highly valued than electricity generation.
- Turbines might cause noise and aesthetic pollution. Wind plants can impact local wildlife. Birds have been killed by flying into spinning turbine blades.



1.7 Wind Machines

Wind machines or wind turbines or wind energy converter, are the devices that convert the wind's kinetic energy into electrical energy. Wind turbines are manufactured in a wide range of sizes, with either horizontal or vertical axes.

Two important wind rotor configurations are as follows:

1. Vertical-axis wind turbines (VAWT), here the axis of rotation is vertical with respect to the ground (and roughly perpendicular to the wind stream). The following are the two main types of VAWT:
 - Darrieus (which uses lift forces generated by aerofoil)
 - Savonius (which uses drag forces)
2. Horizontal-axis wind turbines (HAWT), in which the axis of rotation is horizontal with respect to the ground (and roughly parallel to the wind stream). HAWT can be further divided into three types:
 - Dutch windmills
 - Multi-blade water-pumping windmills
 - High-speed propeller-type wind machines

1.8 Vertical Axis Wind Mills/Turbines

1.8.1 Vertical axis Darrieus wind turbine

- **Constructional details**
- **Tower:** The tower is a hollow vertical rotor shaft, which rotates freely about vertical axis between top and bottom bearings. It is installed above a support structure. The upper part of the tower is supported by guy ropes. The height of the tower of a large turbine is around 100 m

- **Blades:** It has two or three thin, curved blades shaped like an eggbeater in profile, with blades curved in a form that minimizes the bending stress caused by centrifugal forces—the so-called ‘Troposkien’ profile. The blades have airfoil cross section with constant chord length. The pitch of the blades cannot be changed.
- **Support Structure:** Support structure is provided at the ground to support the weight of the rotor. Gearbox, generator, brakes, electrical switchgear and controls are housed within this structure.
- The turbine consists of a number of aero foil usually but not always vertically mounted on a rotating shaft or framework
- Darrieus design, the aero foils are arranged so that they are symmetrical and have zero rigging angles, that is, the angle that the aero foils are set relative to the structure on which they are mounted.
- This arrangement is equally effective no matter which direction the wind is blowing—in contrast to the conventional type, which must be rotated to face into the wind.

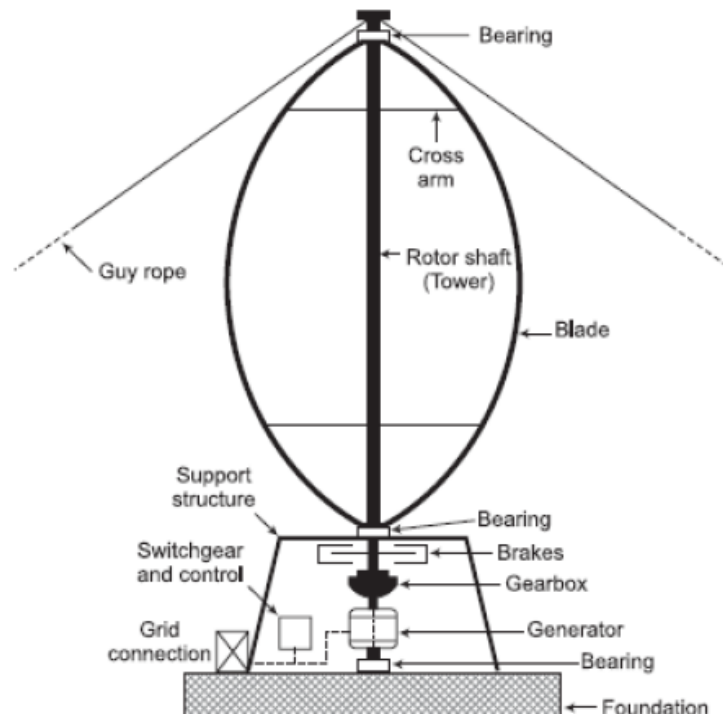


Fig. 1.1 Vertical axis Darrieus wind turbine



1.8.2 Working-Principle

- In terms of operation, Darrieus utilizes the “lift” aerodynamic force to rotate.
- By flowing around the structure, the wind creates suction on the front side of the turbine, driving the wings to rotate.
- Because of the shape of the wings, they do not experience as much drag as Savonius turbines do. Once the rotation starts, Darrieus wind turbines are able to accelerate to rotate faster than the wind speed.
- As the turbine tends to rotate, the electrical generator generates the electrical energy from the mechanical energy supplied by the rotation of rotor shaft.

1.8.3 Advantages

- The equipment (gear box and generator) can be placed close to the ground.
- There is no need of a mechanism to turn the rotor against the wind

1.8.4 Disadvantages

- The efficiency is not very remarkable
- The Darrieus is not a self-starting turbine, the starting torque is very low but it can be reduced by using three or more blades that result in a high solidity for the rotor.
- Because wind speeds are close to the ground level, there is very low wind speed on the lower part of the rotor.
- They are very difficult to mount high on a tower to capture the high level winds. Because of this, they are usually forced to accept the low, more turbulent winds, and they produce less in possibly more damaging winds.

1.9 Vertical-Axis-Savonius-Wind-Turbine

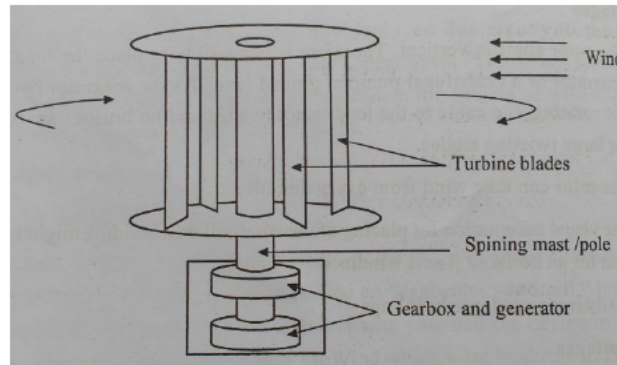


Fig. 1.2 Vertical axis Savonius wind turbine

• Constructional details

- The Savonius wind turbine is a simple vertical axis device having a shape of half-cylindrical parts attached to the opposite sides of a vertical shaft (for two-bladed arrangement) and operate on the drag force, so it can't rotate faster than the wind speed.
- Aerodynamically, it is a drag-type device consisting of two or three scoops.
- Because of the curvature, the scoops experience less drag when moving against the wind than with the wind.

1.9.1 Working-Principle

- The differential drag causes the Savonius turbine to spin.
- Because they are drag-type devices, Savonius turbines extract much less of the wind's power than other similarly sized lift-type turbines.
- As the wind blows into the structure and comes into contact with the opposite faced surfaces (one convex and other concave), two different forces (drag and lift) are exerted on those two surfaces.
- The basic principle is based on the difference of the drag force between the convex and the concave parts of the rotor blades when they rotate around a vertical shaft. Thus, drag force is the main driving force of the Savonius rotor

1.9.2 Advantages

- Always self-starting, if there are at least three scoops
- Relatively easy to make

1.9.3 Disadvantages

- Low efficiency: around 15%.

1.10 Horizontal Axis Wind Turbine

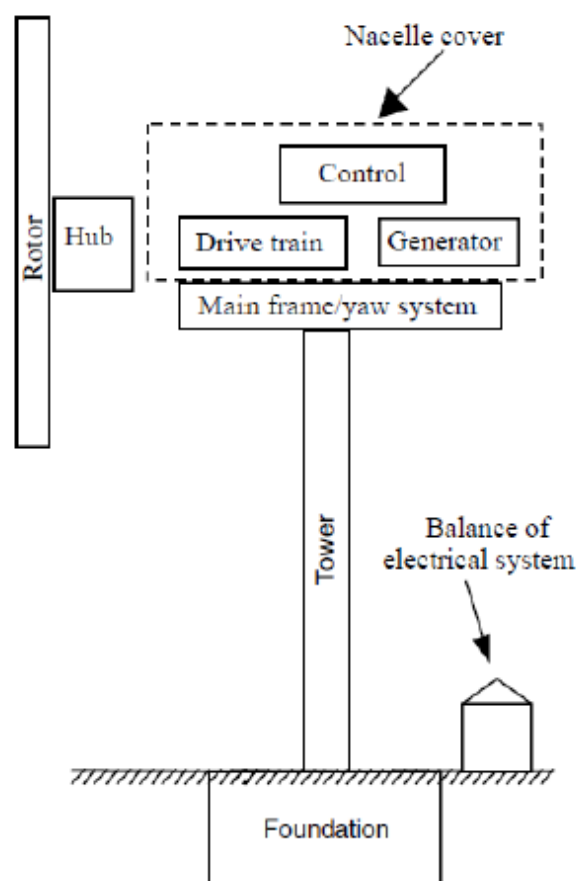


Fig. 1.3 Horizontal axis wind turbine



1.10.1 Constructional details

- The rotor consists of the hub and blades of the wind turbine. Most turbines today have upwind rotors with three blades
- The drive train consists of the other rotating parts of the wind turbine downstream of the rotor. These typically include a low-speed shaft (on the rotor side), a gearbox, and a high-speed shaft (on the generator side). Other drive train components include the support bearings, one or more couplings, a brake
- The purpose of the gearbox is to speed up the rate of rotation of the rotor from a low value (tens of rpm) to a rate suitable for driving a standard generator (hundreds or thousands of rpm).
- This category includes the wind turbine housing, the machine bedplate or main frame, and the yaw orientation system. The main frame provides for the mounting and proper alignment of the drive train components. The nacelle cover protects the contents from the weather
- This category includes the tower itself and the supporting foundation. The principal types of tower design currently in use are the free-standing type using steel tubes, lattice (or truss) towers, and concrete towers.
- A wind turbine control system includes the following components: Sensors – speed, position, flow, temperature, current, voltage, etc.;

Controllers – mechanical mechanisms, electrical circuits;

Power amplifiers – switches, electrical amplifiers, hydraulic pumps, and valves; Actuators – motors, pistons, magnets, and solenoids;

Intelligence – computers, microprocessors.

- In addition to the generator, the wind turbine system utilizes a number of other electrical components. Some examples are cables, switchgear, transformers, power electronic converters, power factor correction capacitors, yaw and pitch motors that forms the balance of electrical system



1.10.2 Working principle

- The horizontal-axis wind turbine (HAWT) is a wind turbine in which the main rotor shaft is pointed in the direction of the wind to extract power
- The rotor receives energy from the wind and produces a torque on a low-speed shaft.
- The low-speed shaft transfers the energy to a gearbox, high-speed shaft, and generator, which are enclosed in the nacelle for protection.
- The low-speed shaft connects to the gearbox, which has a set of gears that increase the output speed of the shaft to approximately 1,800 rpm for an output frequency of 60 Hz (or a speed of 1,500 rpm if the frequency is 50 Hz).
- The high-speed shaft is then connected to the generator, which converts the rotational motion to AC voltage.

1.10.3 Advantages

- High power output
- High efficiency
- Highly reliable
- High operational wind speed

1.10.4 Disadvantages

- Difficult to transport, install and maintenance
- Stronger impact on environment
- Strict regulations to be followed

1.11 Elementary Design Principles

Wind turbine design is the process of defining the form and specifications of a wind turbine to extract energy from the wind. A wind turbine installation consists of the necessary systems needed to capture the wind's energy, point the turbine into the wind, convert mechanical rotation into electrical power, and other systems to start, stop, and control the turbine.

- **Aerodynamics:** The shape and dimensions of the blades of the wind turbine are determined by the aerodynamic performance required to efficiently extract energy from the wind, and by the strength required to resist the forces on the blade.
- **Power control:** The centrifugal force on the spinning blades increases as the square of the rotation speed, which makes this structure sensitive to over speed. A wind turbine is designed to produce power over a range of wind speeds. The cut-in speed is around 3–4 m/s for most turbines, and cut-out at 25 m/s. A control system involves three basic elements: sensors to measure process variables, actuators to manipulate energy capture and component loading, and control algorithms to coordinate the actuators based on information gathered by the sensors.
- **Stall:** A stall on an airfoil occurs when air passes over it in such a way that the generation of lift rapidly decreases. Usually this is due to a high angle of attack (AOA), but can also result from dynamic effects. The blades of a wind turbine with fixed pitch can be aerodynamically designed to stall in high wind speeds, causing slower rotation.
- **Furling:** Furling works by decreasing the angle of attack, which reduces the induced drag from the lift of the rotor, as well as the cross-section. A fully furled turbine blade, when stopped, has the edge of the blade facing into the wind.
- **Yawing:** Modern large wind turbines are typically actively controlled to face the wind direction measured by a wind vane situated on the back of the nacelle. By minimizing the yaw angle (the misalignment between wind and turbine pointing direction), the power output is maximized and non-symmetrical loads minimized
- **Turbine size:** For a given survivable wind speed, the mass of a turbine is approximately proportional to the cube of its blade-length. Wind power intercepted by the turbine is proportional to the square of its blade-length. The maximum blade-length of a turbine is limited by both the strength, the stiffness of its material, and transportation considerations.

1.12 Co-efficient of Performance of A Wind Mill Rotor

- It is the proportion of the power in the wind that the rotor can extract (it is also called power co-efficient or efficiency; symbol C_p) and its variation as a function of tip-speed ratio is commonly used to characterize different types of rotor.
- It is physically impossible to extract all the energy from the wind, without bringing the air behind the rotor to a standstill.
- Consequently, there is a maximum value of C_p of 59.3% (known as the Betz limit), although in practice, real wind rotors have maximum C_p values in the range of 25%-45%.
- The performance coefficient of a rotor is the fraction of wind energy passing through the rotor disc, which is converted into shaft power. This is a measure of the efficiency of the rotor and it varies with the tip-speed ratio.

1.13 Aerodynamic Considerations In Wind Mill Design

The VAWT and HAWT use either lift or drag forces to harness the wind. Out of these types, the horizontal- axis lift device is the most commonly used. In fact, other than a few experimental machines, virtually all windmills come under this category.

There are two primary physical principles by which energy can be extracted from the wind. These are through the creation of either lift or drag force (or through a combination of the two), as shown in the below Fig. 1.4

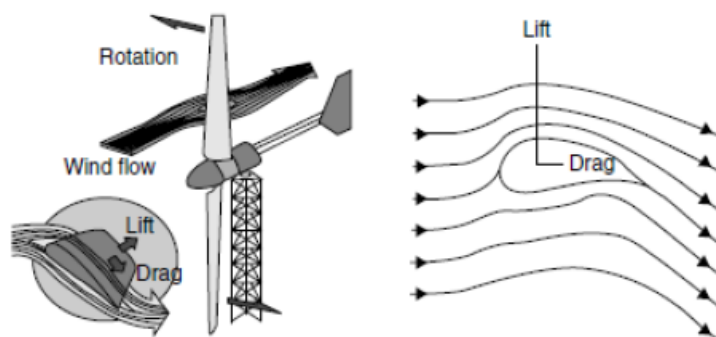


Fig. 1.4 Horizontal axis wind turbine

Air flow over a stationary airfoil produces two forces, a lift force perpendicular to the air flow and a drag force in the direction of air flow, as shown in the above figure.

The existence of the lift force depends upon laminar flow over the airfoil, which means that the air flows smoothly over both sides of the airfoil. If turbulent flow exists rather than laminar flow, there will be little or no lift force.

The air flowing over the top of the airfoil has to speed up because of a greater distance to travel and this increase in speed causes a slight decrease in pressure. This pressure difference across the airfoil yields the lift force, which is perpendicular to the direction of air flow. The air moving over the airfoil also produces a drag force in the direction of the air flow. This is a loss term and is minimized as much as possible in high- performance wind turbines

- **Lift Force:** The lift force (FL) arises in a direction that is perpendicular to the airstream caused by Bernoulli's effect that lowers the pressure on the top of the airfoil when compared with the pressure on its bottom. The curvature on the top leads to a higher stream velocity than at the bottom and hence a lower pressure.

Let (FL) is the lift force in Newton, (SL) is the cross-sectional area of airfoil in m², ρ is the air density in kg/m³, and V is the wind speed in m/s. Then, lift coefficient (CL) is defined as follows

$$C_L = [F_L / S_L] / [(1/2) \rho V^2]$$

- **Drag force (FD):** It is described as follows, Where CD = drag coefficient and SD = Effective area of airfoil in the direction of drag force.

$$C_D = [F_D / S_D] / [(1/2) \rho V^2]$$

- The lift and drag force vary with the angle that rotor blade makes with the direction of wind stream. This angle is called as angle of attack. The resultant of drag and lift forces constitute the thrust force that effectively rotate the blade.

1.14 Numerical

A three-bladed wind rotor with blade length of 52 m is operating in a wind stream having wind velocity of 12 m/s. Air density is 1.23 kg/m³ and power coefficient may be taken as 0.4. Calculate the extractable power from the wind.

Solution:

Given data are as follows:

Blade length, $L = 52$ m; wind speed, $v = 12$ m/s; air density, $\rho = 1.23$ kg/m³; power coefficient, $C_p = 0.4$.

Thus, $A = \text{swept area} = \pi r^2 = \pi (52)^2 = 8495 \text{ m}^2$

$$P_{\text{Available}} = C_p \left(\frac{1}{2} \rho A v^3 \right) = 0.4 \times \frac{1}{2} \times 1.23 \times 8495 \times (12)^3 = 3.6 \text{ MW}$$

1.15 Geothermal Energy Conversion

1.16 Introduction

Geothermal energy originates from earth's interior in the form of heat. Volcanoes, geysers, hot springs and boiling mud pots are visible evidence of the great reservoirs of heat that lies within earth. Although the amount of thermal energy within the earth is very large, useful geothermal energy is limited to certain sites only, as it is not feasible to access and extract heat from a very deep location. The sites where it is available near the surface and is relatively more concentrated, its extraction and use may be considered feasible. These sites are known as geothermal fields. As per US Geological Survey, the entire heat content of the earth's crust up to a depth of 10 km above 15 °C is defined as geothermal resource. As such the geothermal resource is estimated to be more than 2.11×10^{25} J, which is equivalent to 109 MTOE (million tons of oil equivalent). This is a huge amount of energy, enough to supply our energy needs at current rates for 3,50,000 years. Thus it is considered an inexhaustible and renewable source. However, it is a low-grade thermal energy form and its economic recovery is not feasible everywhere on the surface of the earth. Practically it is not the size of the resource that limits its use but the availability of technology that can tap the resource in an economic manner.

Low temperature resources, i.e. “geysers” have been used from time immemorial for applications such as therapeutic hot baths, cooking, space and water heating. Most geothermal resources produce low-grade heat at about 50–70 °C, which can be used directly for thermal applications. Occasionally, geothermal heat is available at temperatures above about 90 °C, and so electrical power production from turbines can be contemplated. World’s total present (end of year 2012) installed electrical power generating capacity from geothermal resource is about 11,490 MWe and direct thermal use installed capacity is 50,000 MWt. Globally, geothermal power is growing steadily at a rate of about 5 per cent per year. The annual growth in energy output over the past five years has been 3.8 per cent for electricity production and around 10 per cent for direct use (including geothermal heat pumps. The geothermal electrical energy production in El Salvador is 25 per cent of the country’s total production. At present the capital cost of a geothermal power plant is about Rs. 150–375 crore (Euro 2–5 million) per MWe of electric capacity. The geothermal electricity cost is about Rs 3–7.5 (Euro 0.01 to 0.1) per kWh.

1.17 Types Of Geothermal Resources

There are four types of geothermal resources:

- (i) hydrothermal,
- (ii) Geo-pressured,
- (iii) Hot dry rock (HDR)
- (iv) magma.

1. At present the technology for economic recovery of energy is available for hydrothermal resource only. Thus this is the only commercially used resource at present. Other resources are going through development phase and have not become commercial so far.

Hydrothermal Resources Hydrothermal resources arise when underground water has access to high temperature porous rocks, capped by a layer of solid impervious rock. Thus water is trapped in the underground reservoir (aquifers) and is heated by surrounding rocks. Heat is supplied by magma by upward conduction through solid rocks below the reservoir. Thus it forms a giant underground boiler. Under high pressure, the temperature can reach as high as 350 °C. The hot water often escapes through fissures in the rock, thus

forming hot springs or geysers. Sometimes steam escapes through cracks in the surface. These are called fumaroles. In order to utilize the hydrothermal energy, wells are drilled either to intercept a fissure or, more commonly into the hydrothermal reservoir as shown in Fig. 1.5.

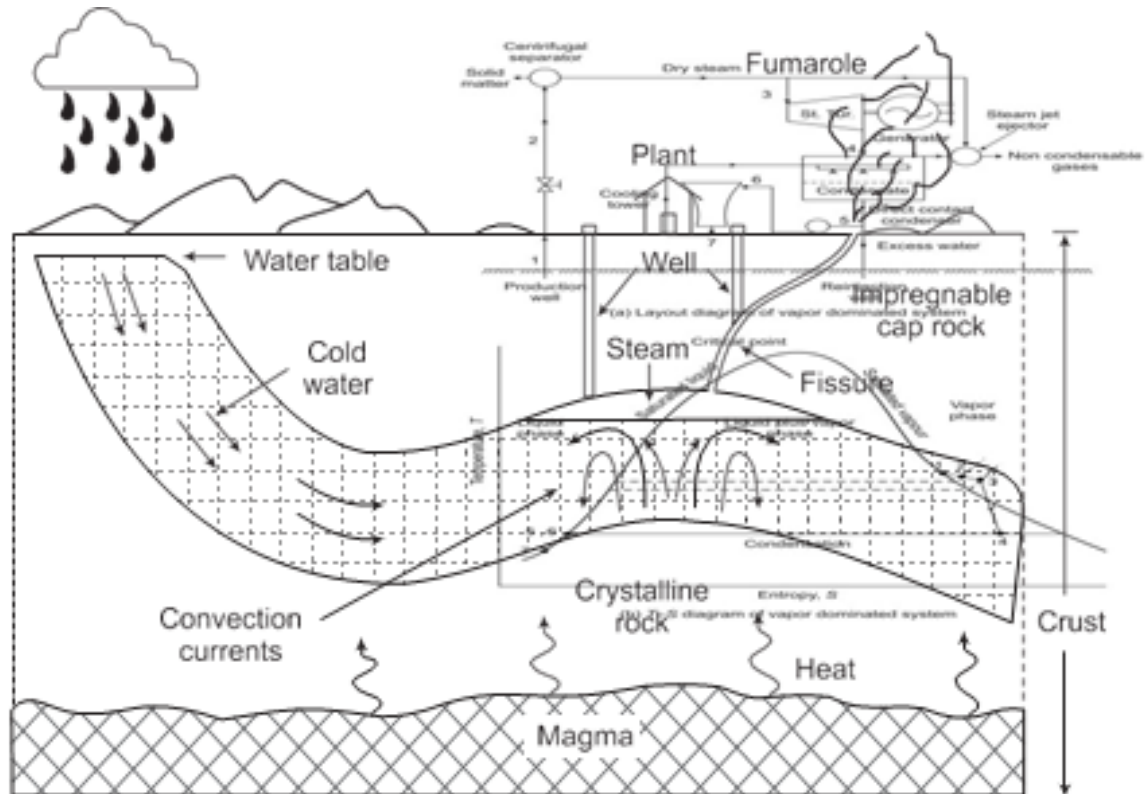


Fig. 1.5 General simplified structure of hydrothermal resource

The hydrothermal resources are located at shallow to moderate depths (from approximately 100 m to 4,500 m). Temperatures for hydrothermal reserves used for electricity generation range from 90 °C to 350 °C but roughly two-thirds are estimated to be in the moderate temperature range (150 °C to 200 °C). For practical purposes, hydrothermal resources are further subdivided into and

For practical purposes, hydrothermal resources are further subdivided into

- (i) vapour dominated (dry steam fields), (ii) liquid dominated (wet steam fields)

hot water resource. Vapour dominated fields deliver steam with little or no water and liquid dominated fields produce mixture of steam and hot water or hot water only. The system to utilize the energy depends on the type of resource.

Vapour Dominated (Dry Steam) System

Dry steam fields occur when the pressure is not much above the atmospheric pressure and the temperature is high. Water boils underground and generates steam. The most important known dry steam fields are:

- (a) “The geysers” regions in California, which may be the largest,
- (b) the Larderello and some smaller areas in Italy and
- (c) a small field (or fields) at Matsukawa, Japan. Dry steam field at Larderello is especially ideal because its wells produce virtually pure steam with no water. The layout of vapor-dominated system is shown in Fig. ?? The operation is explained with the help of T-S diagram given in Fig. ?. The steam is extracted from the well (1) where it is nearly saturated. The extracted steam is then cleaned in centrifugal separator to remove solid matter. While passing through the well, as well as centrifugal separator the pressure drops, which causes it to slightly super heat. The steam is then supplied to a turbine (3) at temperature of about 165 °C and pressure of about 7.8 atm. (the temperature and pressure in the reservoir are higher) and allowed to expand (4). The exhaust steam of turbine is condensed in direct contact condenser, in which the steam is condensed by direct contact with cooling water. The resulting warm water (5) is circulated and cooled in cooling tower and returned to the condenser (7). The condensation of steam continuously increases the volume of cooling water. Excess water is reinjected at some distance deep into the ground for disposal. The non- condensable gases are removed from the condenser by steam jet ejection.

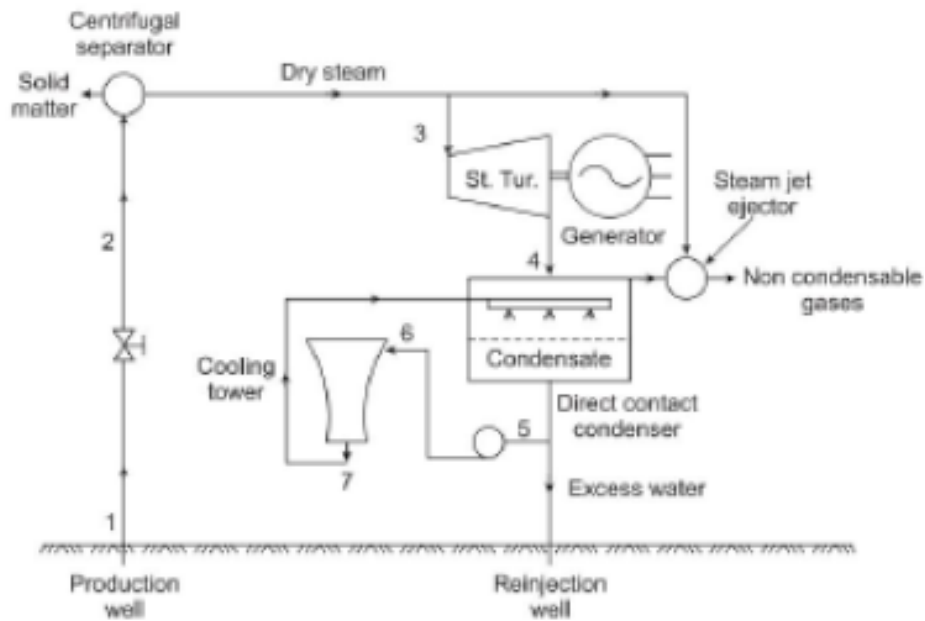
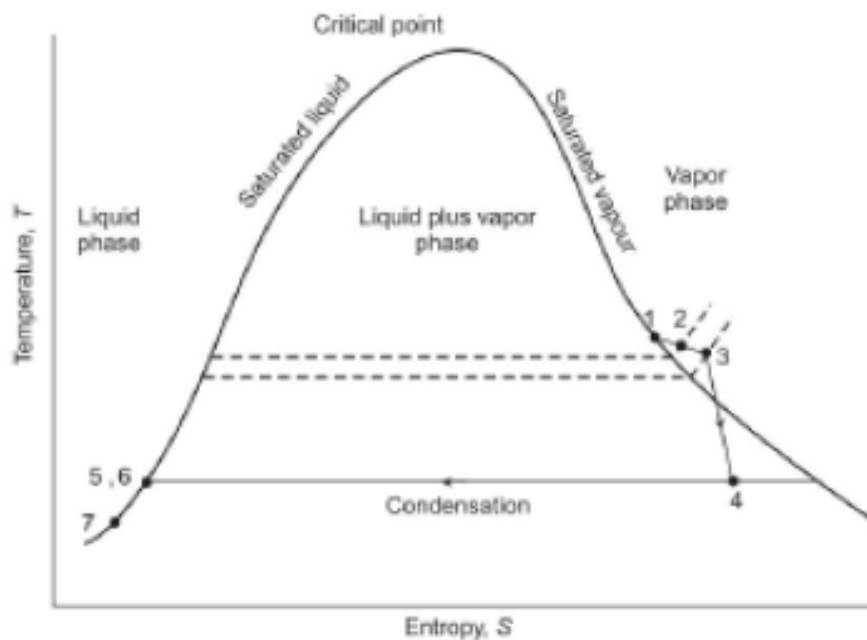


Fig. 1.6 Schematic diagram of Vapour Dominated (Dry Steam) System



(b) T-S diagram of vapor dominated system

Fig. 1.7 T-S Vapour Dominated (Dry Steam) System

1.18 Liquid Dominated (Wet Steam) System

Steam plants are the most cost-effective technology, when the resource temperature is above about 175 °C. Therefore, liquid dominated or wet steam fields may further be subdivided into (a) high temperature (above 175 °C), where steam plants can be used and (b) low temperature (below 175 °C) fields where other technologies are used. **Liquid Dominated – High Temperature System** In high temperature liquid dominated reservoir, water temperature is above 175 °C; however, it is under high pressure and remains in liquid state. The most extensive development has been in the volcanic Wairakei field in New Zealand, the first of its kind, was built in 1958. In this field the reservoir temperature and pressures are 230 °C and 40 atm respectively and depths are 600 m to 1400 m. When water is brought to the surface and pressure is reduced, rapid boiling occurs and it “flashes” into steam and hot water. The operation of single flash system is illustrated in Fig. 1.8.

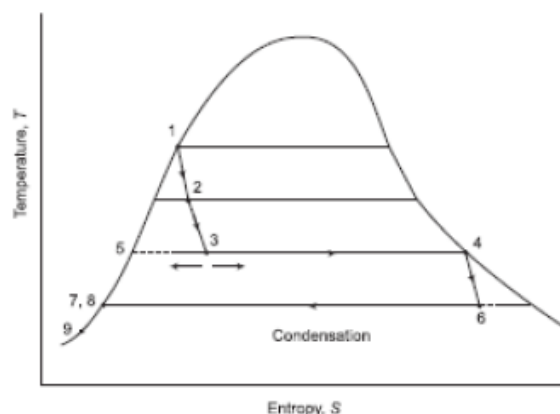
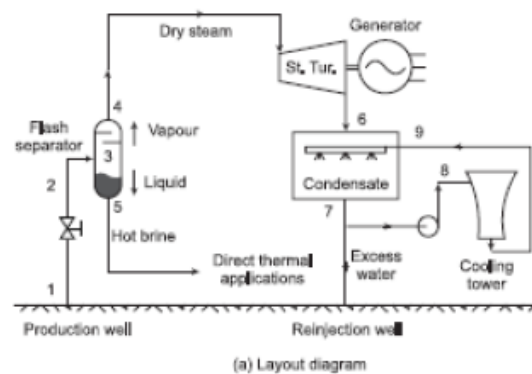


Fig. 1.8 Single flash, wet steam high temperature hydrothermal system

The hot water from the well (1) is brought to well head (2) at a lower pressure. Process 1–2 is essentially a constant enthalpy throttling process that results in a two-phase mixture of low quality at 2. This is throttles further in a flash separator (3) to get a two-phase mixture of slightly higher quality. This mixture is now separated into dry saturated steam at 4 and saturated brine at 5. The highly saline hot water (known as brine) can be used for direct heat and then reinjected into the ground. The separated steam at 4 is expanded in a turbine to 6. The maximum initial steam temperature is about 175 °C and the gauge pressure is 3.5 atm. The exhaust steam at 6 is mixed with cooling water in a direct contact condenser and the resulting mixture at 7 is directed to a cooling tower.

In dual flash systems,

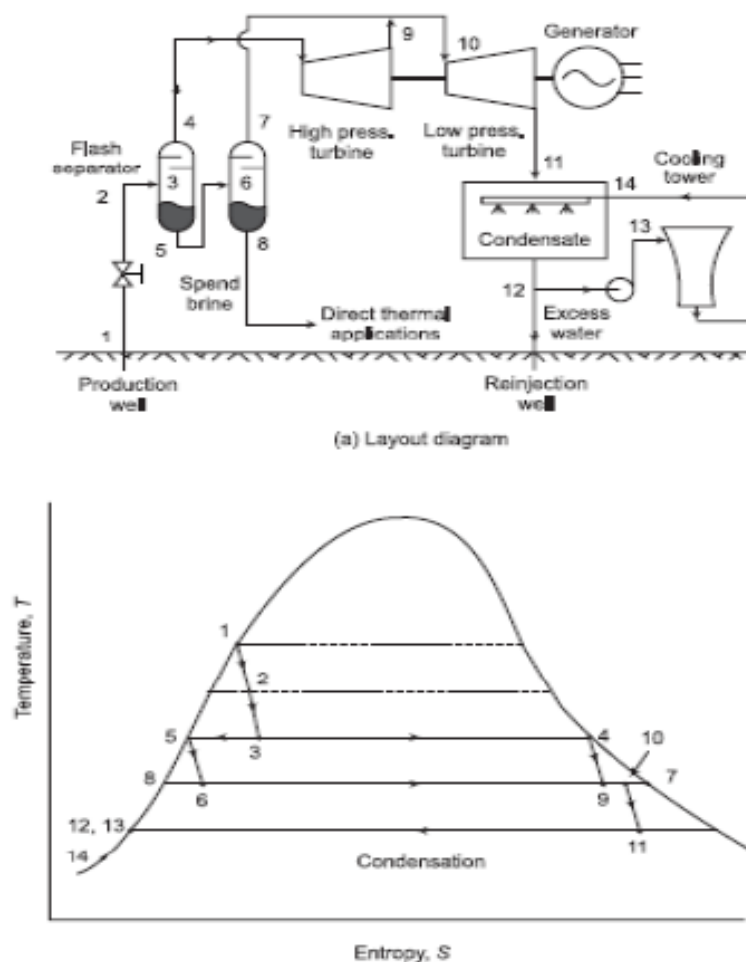


Fig. 1.9 Double flash, wet steam high temperature hydrothermal system

In dual flash systems, the steam is flashed a second time from the remaining hot fluid of the first stage, separated and fed into dual inlet turbine or into two separate turbines. The efficiency of such a plant is around 8 per cent. The layout of double flashed systems along with T-S diagrams is shown in Fig. 1.9. The 50 MWe Hatchobane plant, Kyushu in Japan is a double flash system. As of 2013, the largest liquid system is Cerro Prieto in Mexico, which generates 750 MWe from temperatures reaching 350 °C (662 °F). The Salton Sea field in Southern California offers the potential of generating 2000 MWe. Such systems are also being implemented in Philippines and other places.

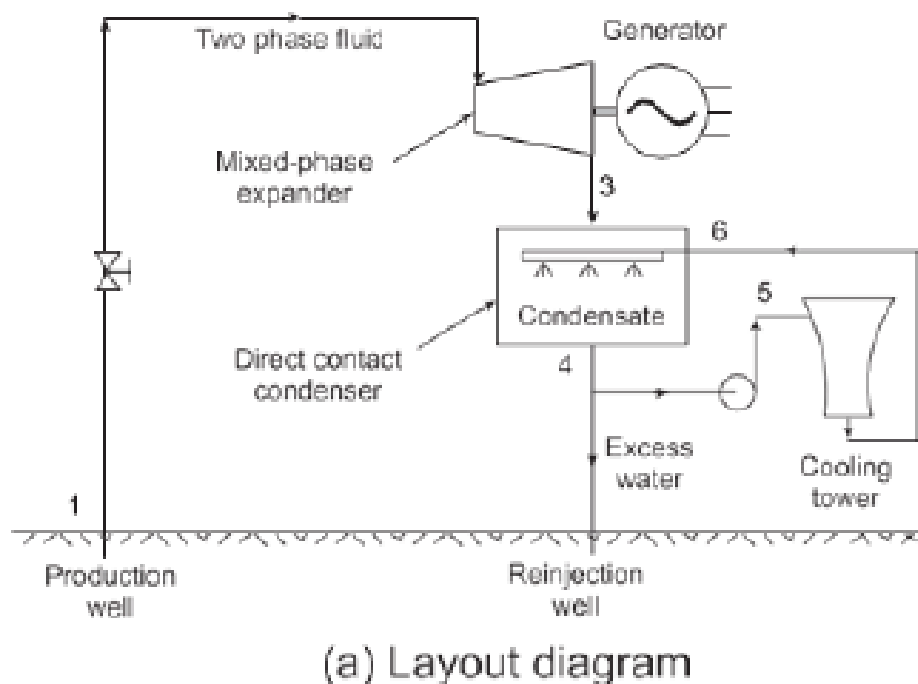


Fig. 1.10 Double flash, wet steam high temperature hydrothermal system

A third approach is called Total Flow Concept, in which a special mixed phase turbine is used that would utilize both the kinetic energy as well as heat energy of the two-phase (steam–liquid) mixture produced by flashing the geothermal brine. The overall efficiency in such a system should be greater than other methods where only heat content of the brine is utilized. A layout and corresponding T-S diagram of total flow power plant is given in Fig. 9.7. The hot fluid from geothermal well (1) is throttled to get a two-phase

mixture of low quality at 2. At this point the two phases are not separated, and the total flow is piped to turbine and allowed to expand to the discharge is condensed to 4 in direct contact condenser. The excess brine is reinjected to ground.

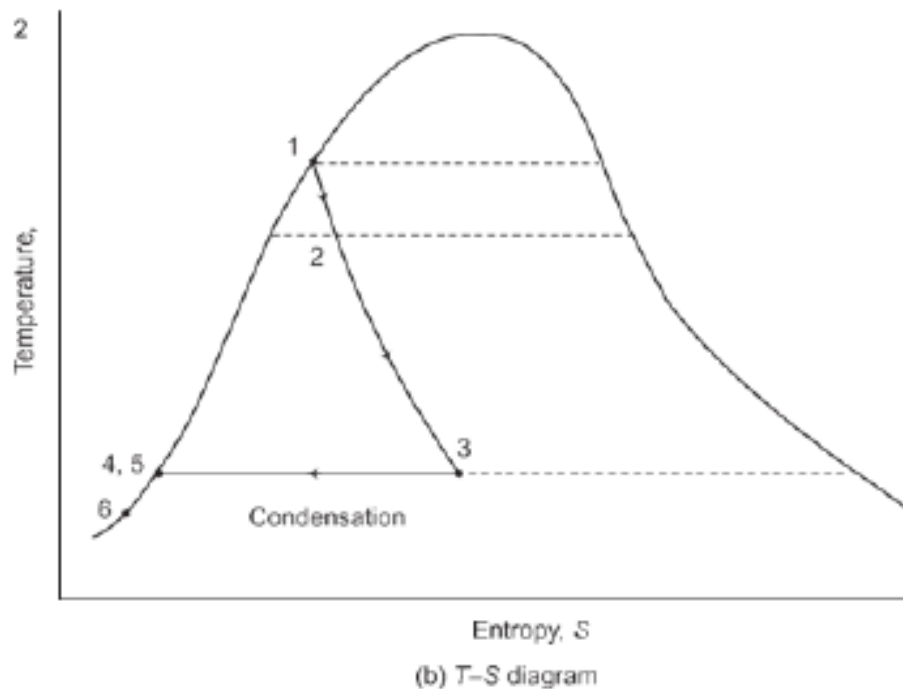


Fig. 1.11 Double flash, wet steam high temperature hydrothermal system

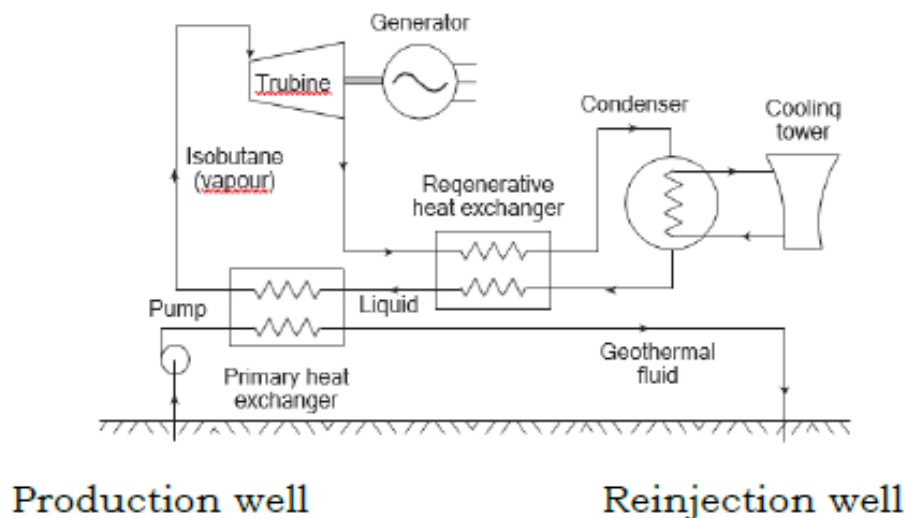
Liquid Dominated – Low Temperature System These resources are available at moderate temperature range of 90 °C – 175 °C. They are common in extensional terrains, where heating takes place via deep circulation along faults, such as in the Western US and Turkey. This temperature is not enough for efficient flash steam production and requires pumping. A binary-fluid system is employed, where the heat of geothermal fluid is used to vapourize a volatile organic fluid, such as isobutene (B.P. 10 °C), under pressure in a primary heat exchanger. The geothermal fluid is reinjected after extraction of heat. This vapourized fluid serves as working fluid in a Rankine cycle plant. The exhaust vapour from the turbine is cooled in the regenerative heat exchanger and then condensed in a condenser. The condensed liquid isobutene is returned to primary heat exchanger by way of the regenerative heat exchanger. These binary plants originated in the Soviet Union in

late 1960s and predominate in new US plants. The layout of the system is shown in Fig. 9.8. Binary plants have no emissions.

The thermal efficiency of such a plant is typically about 10–13 per cent. These plants do not produce any steam condensate and have to rely on external source of cooling water or air-cooling.

Such plants are in use at East Mesa, Mammoth Lake in California and other places in USA, Kawerau, New Zealand and many other places in the world. Presently some 1790 MWe binary power plants are in operation worldwide [52].

Main advantages of binary systems are: (i) they almost avoid corrosion, scaling and environmental problems as the geothermal fluid circulates through a closed- cycle and all the fluid is reinjected and (ii) in many cases, they are capable of higher conversion efficiencies than flash steam plants.



1.19 Geo-pressured Resources

Hydrothermal reservoirs of low to moderate temperatures (20 °C–150 °C) can be used to provide direct heat for residential and industrial uses. The hot water is brought to the surface where a heat exchanger system transfers its heat to another fluid (liquid or

air); although the resource can be used directly if the salt and solid content is low. The geothermal fluid is reinjected into the ground after extraction of heat. The heated fluid transports heat to the place of use. Recent surveys have identified a large potential for direct use geothermal applications. Iceland is the world leader in direct applications. Some 92.5 per cent of its homes are heated with geothermal energy, saving Iceland over \$100 million annually in avoided oil imports. Reykjavík, Iceland has the world's biggest district heating system. Once known as the most polluted city in the world, it is now one of the cleanest.

Energy of hot water resource can also be utilized in a hybrid system consisting of geothermal–conventional thermal (fossil fuel or biomass based) system. In this system, hot water resource is used to preheat feed water and / or air for combustion. Geothermal heat replaces some or all of the feed water heaters, depending upon its temperature. A 30 MWe geothermal-wood waste hybrid plant is in operation at Honey Lake, California since 1989.

1.20 Hot Dry Rock Resources

While drilling for oil and gas, hot salty water (brine) reservoirs, at moderately high temperature, (90 °C to 200 °C), and under great pressure are found at a depth of 3 to 6 km. Because of the very high pressure of the water, up to 1350 atm in the deepest layer, these reservoirs are referred to as geo-pressured. A special feature of geo- pressured water is that it also contains a significant amount of dissolved methane gas, usually

1.9 – 3.8 m³ per m³ of water. The solubility of methane in water at normal pressure is quite low, but it increases with pressure. When the water is brought to the surface and its pressure reduced, the methane gas is released from the solution. Thus methane can be extracted from brine by simple and economical gravity separation technique. This resource is potentially very promising because three types of energy can be extracted from the wells: (i) thermal energy from the heated fluids, (ii) mechanical (hydraulic) energy from the high pressures involved and (iii) chemical energy from burning of methane gas.

The extent of geo-pressured reserves is not yet well known worldwide, and the only major resource area identified to date is along the Texas and Louisiana coast of the Gulf of Mexico. The belt is about 1200 km in length along the coast and extends up to about 160 km inland and 240 km offshore 1.12.

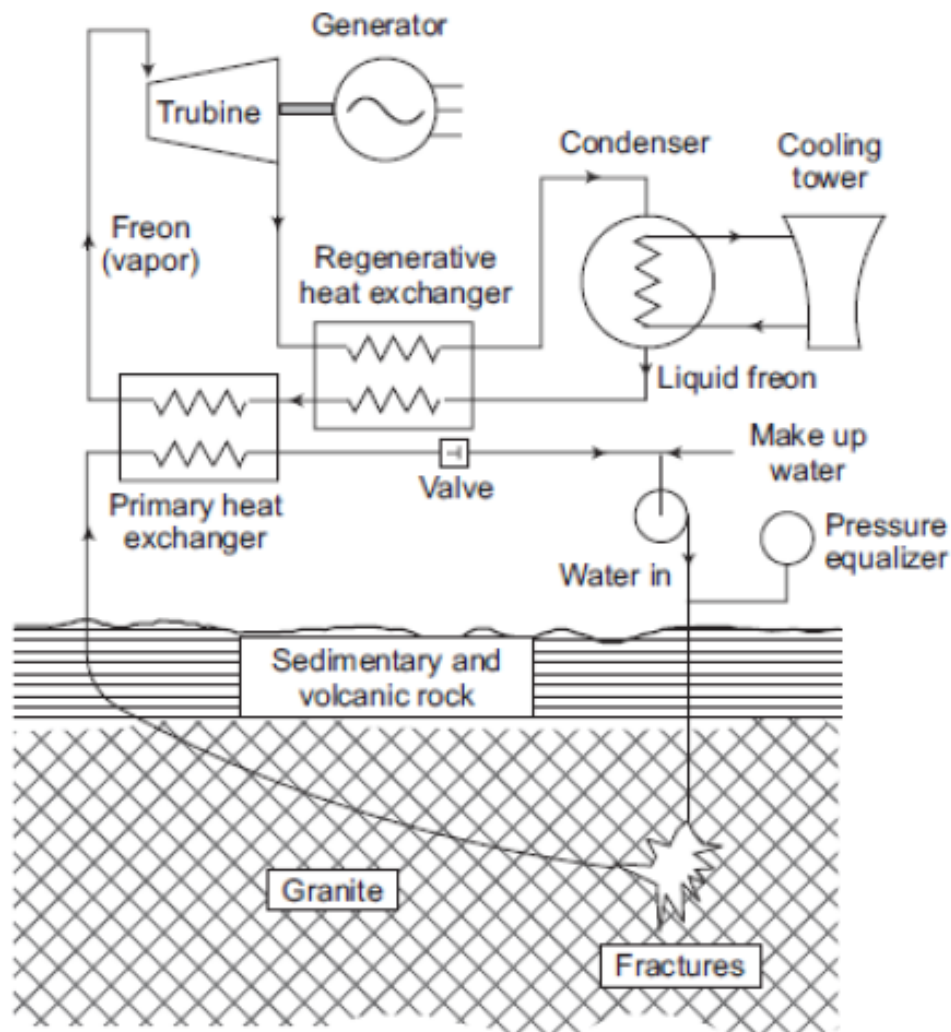


Fig. 1.12 Schematic Hot Dry Rock Resources

The potential of geo-pressured resource of this area has been estimated as 23 to 240 GWe for 30 years. Experimental wells are already in place in Texas and Louisiana. A hybrid geo-pressured conversion unit (1 MWe) was successfully operated in 1990 as a demonstration unit. Studies are underway to identify reservoirs with sufficient volume and permeability to sustain large water flows over long time periods and the extent of dissolved methane gas.

1.21 Magma Resources

At some places, molten or partially molten rock (magma chamber), at temperatures 650 °C to 1,200 °C, occurs at depths 5 km – 10 km. These resources are located, especially in the vicinity of recent volcanic activity (e.g. Hawaii). Very high temperature and large volume make magma a huge potential energy source, the largest of all geothermal resources. However, successful magma drilling technology has not been established yet. Extracting magma energy is expected to be the most difficult of all resource types. Magma technology will require special drilling technology to deal with the interaction of the drill bit with molten rock, the effects of dissolved gases, and mechanisms of heat transport in molten magma. This resource has not been developed as yet.

1.22 Geothermal Energy In India

Though India has been one of the earliest countries to begin geothermal projects way back in the 1970s, but at present there are no operational geothermal plants in India. There is also no installed geothermal electricity generating capacity as of now and only direct uses (e.g. drying) have been detailed.

A systematic collaborative, research, development and demonstration program is undertaken with different organizations, viz., IIT Delhi, National Aeronautic Limited, Bangalore, Geological Survey of India, National Geophysical Research Institute (NGRI), Hyderabad, Oil & Natural Gas Corporation, etc. As a result of various resource assessment studies/surveys, nearly 350 potential hot springs, distributed in seven geothermal provinces, have been identified throughout the country. These springs are perennial and their surface temperatures range from 37 to 90 °C with a cumulative surface discharge of over 1000 l/min. Most of them are low temperature hot water resources and can best be utilized for direct thermal applications. Only some of them can be considered suitable for electrical power generation. The potential for power generation at these sites has been estimated as about 10,600 MW. The use of geothermal energy has already been demonstrated in the country for small-scale power generation and thermal applications. Small direct heat pilot plants have been installed at Puga and Chumathang (in Ladakh, Jammu and Kashmir) and Manikaran (H.P.), etc.

The seven geothermal provinces include, The Himalayas: Sohana: West coast: Cambay: Son–Narmada–Tapi (SONATA): Godavari and Mahanadi. These provinces are associated with major rifts or subduction tectonics and registered high heat flow and high geothermal gradient. The locations of geothermal fields in the country are shown in Fig. 9.12. The heat flow values and thermal gradients at these locations are

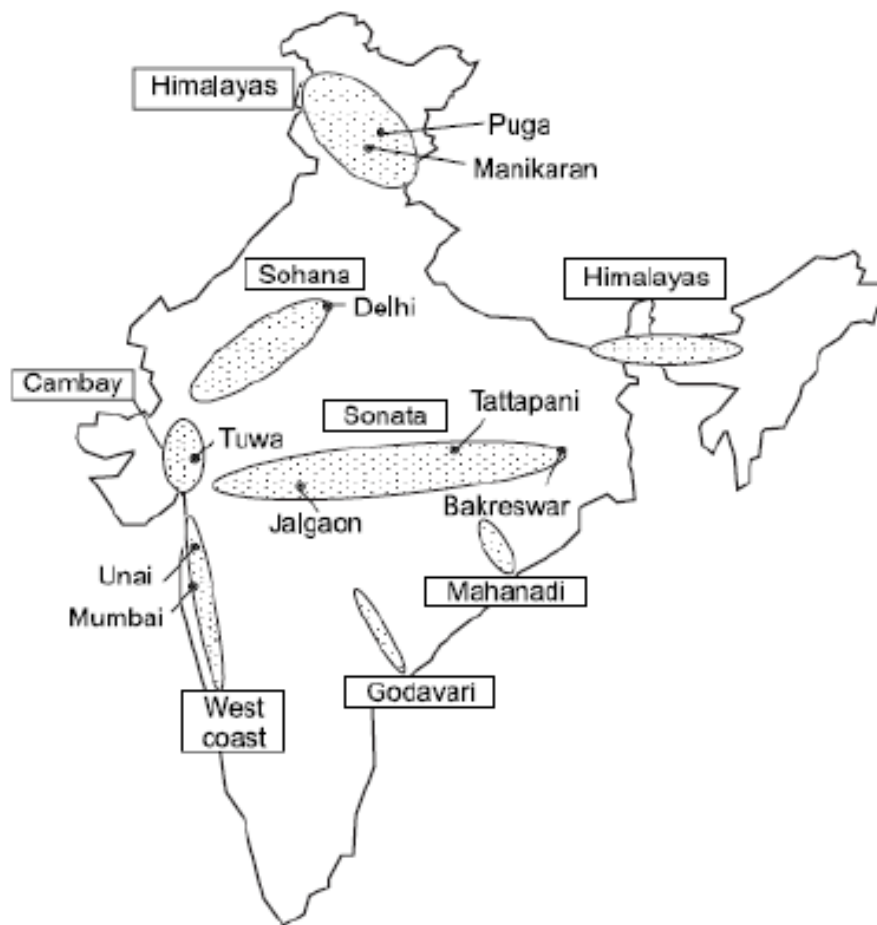


Fig. 1.13 Geothermal provinces of India

Figure (1.13) Geothermal provinces of India (not to scale) The geothermal reservoirs suitable for power generation have been located at Tattapani (in Sarguja district), Chhattisgarh and Puga valley of Ladakh, Jammu & Kashmir. Binary cycle system may be used for power generation from these medium enthalpy resources. A study by National

Geophysical Research Institute (NGRI), Hyderabad has confirmed the presence of a 260 °C hydrothermal reservoir at a depth of 3 km at Tattapani. The Ministry of New and Renewable Energy Sources is planning to develop this field for power generation. Nation Hydroelectric Power Corporation (NHPC) has been entrusted the job of installing a 300 kW demonstration electric power production plant at a cost of Rs. 4.6 crore. Exploratory study is underway by NGRI, Hyderabad at Sutluj-Spiti, Beas and Parbati valley in Himachal Pradesh, Badrinath- Tapovan in Uttaranchal and Surajkund in Jharkhand to explore potential sites for power generation.

S.N.	Geothermal provinces	Main Locations	Estimated reservoir temp.(°C)	Heat flow (mW/m ²)	Thermal gradient (°C/km)
1.	Himalayas	Puga Valley (J&K), Manikaran (H.P.)	260	468	234
2.	Sohana		70		76–96
3.	West coast	Unai (Maharashtra)	102–137	129	59
4.	Cambay	Tuwa (Gujrat)	150–175	93	70
5.	Son–Narmada–Tapi (SONATA)	Jalgaon (Maharashtra), Tattapani (Chhattisgarh), Bakreswar (Bihar)	105–217	120–260	60–90
6.	Godavari		175-215	104	60
7.	Mahanadi		130	200	90

Since majority of these resources are located in rural India, these springs can support small-scale industries in such areas.

1.22.1 Advantages

1. Environmentally Friendly

Geothermal energy is more environmentally friendly than conventional fuel sources such as coal and other fossil fuels. In addition, the carbon footprint of a geothermal power plant is low. While there is some pollution associated with geothermal energy, this is relatively minimal when compared to fossil fuels.



1. Renewable

Geothermal energy is a source of renewable energy that will last until the Earth is destroyed by the sun in around 5 billion years. The hot reservoirs within the Earth are naturally replenished, making it both renewable and sustainable.

1. Huge Potential

Worldwide energy consumption is currently around 15 terawatts, which is far from the total potential energy available from geothermal sources. While we can't currently use most reservoirs there is a hope that the number of exploitable geothermal resources will increase with ongoing research and development in the industry. It is currently estimated that geothermal power plants could provide between 0.0035 and 2 terawatts of power.

1. Sustainable / Stable

Geothermal provides a reliable source of energy as compared to other renewable resources such as wind and solar power. This is because the resource is always available to be tapped into, unlike with wind or solar energy.

1. Heating and Cooling

Effective use of geothermal for electricity generation requires water temperatures of over 150°C to drive turbines. Alternatively, the temperature difference between the surface and a ground source can be used. Due to the ground being more resistant to seasonal heat changes than the air, it can act as a heat sink/ source with a geothermal heat pump just two metres below the surface.

1. Reliable

Energy generated from this resource is easy to calculate since it does not fluctuate in the same way as other energy sources, such as solar and wind. This means we can predict the power output from a geothermal plant with a high degree of accuracy.

1. No Fuel Required



Since geothermal energy is a naturally occurring resource there is no fuel required, such as with fossil fuels that are a finite resource which needs mining or otherwise extracting from the earth.

1. Rapid Evolution

There is a great deal of exploration into geothermal energy at the moment, meaning that new technologies are being created to improve the energy process. There are an increasing number of projects to improve and grow this area of industry. With this rapid evolution many of the current cons of geothermal energy will be mitigated against.

1.22.2 Disadvantages of Geothermal Energy

1. Location Restricted

The largest single disadvantage of geothermal energy is that it is location specific. Geothermal plants need to be built in places where the energy is accessible, which means that some areas are not able to exploit this resource. Of course, this is not a problem if you live in a place where geothermal energy is readily accessible, such as Iceland.

1. Environmental Side Effects

Although geothermal energy does not typically release greenhouse gases, there are many of these gases stored under the Earth's surface which are released into the atmosphere during digging. While these gases are also released into the atmosphere naturally, the rate increases near geothermal plants. However, these gas emissions are still far lower than those associated with fossil fuels.

1. Earthquakes

Geothermal energy also runs the risk of triggering earthquakes. This is due to alterations in the Earth's structure as a result of digging. This problem is more prevalent with enhanced geothermal power plants, which force water into the Earth's crust to open up fissures to greater exploitation of the resource. However, since most geothermal plants are away from population centres, the implications of these earthquakes are relatively minor.



1. High Costs

Geothermal energy is an expensive resource to tap into, with price tags ranging from around \$2-\$7 million for a plant with a 1 megawatt capacity. However, where the upfront costs are high, the outlay can be recouped as part of a long-term investment.

1. Sustainability

In order to maintain the sustainability of geothermal energy fluid needs to be pumped back into the underground reservoirs faster than it is depleted. This means that geothermal energy needs to be properly managed to maintain its sustainability.

It is important for industry to assess the geothermal energy pros and cons in order to take account of the advantages while militating against any potential problems.